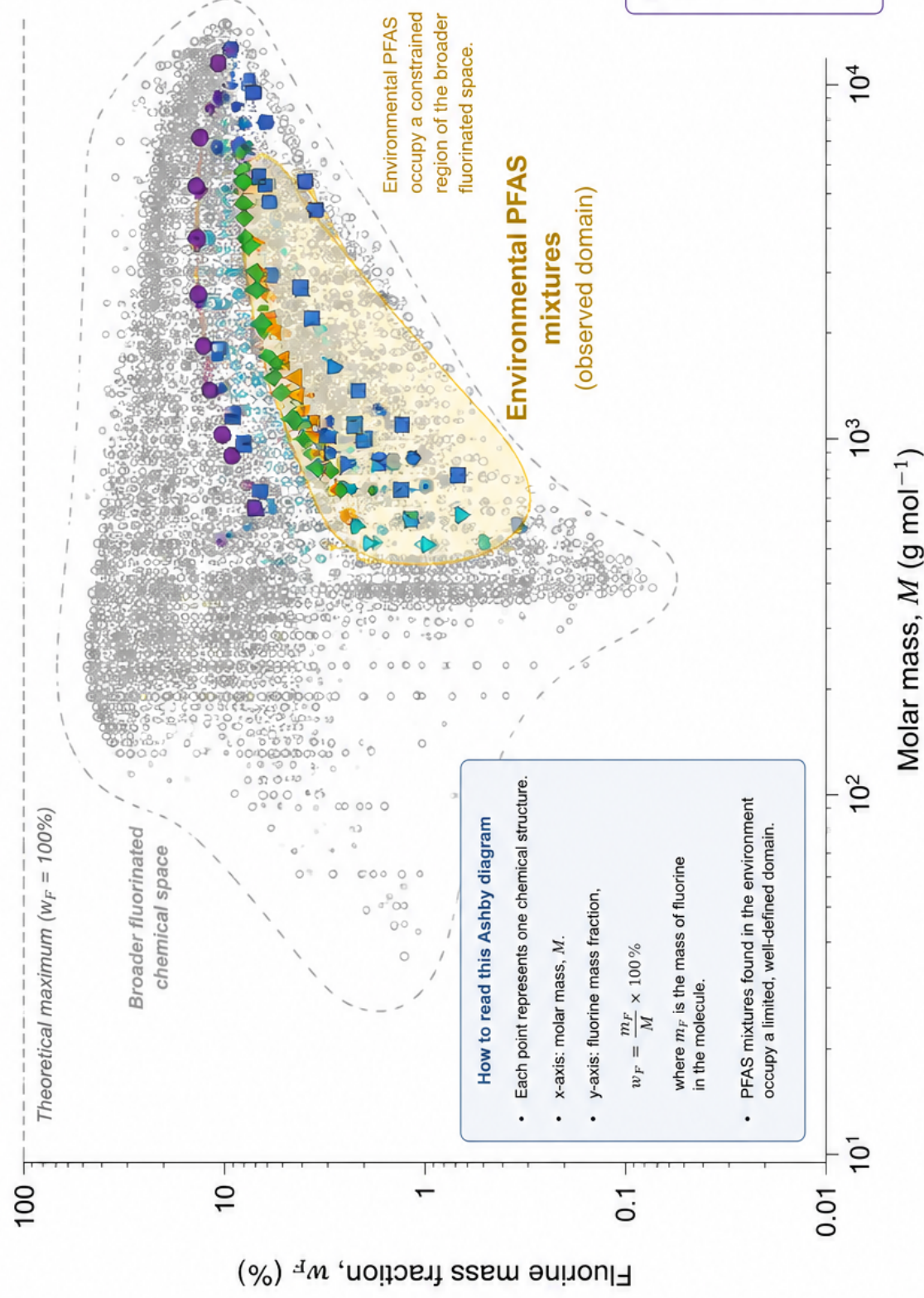




The grey cloud represents ~1.8 million fluorinated organic compounds from PubChem. Colored symbols indicate major PFAS classes detected in environmental monitoring. The highlighted domain corresponds to the composition of PFAS mixtures observed in more than 800,000 environmental samples from Europe and North America (this study). Environmental PFAS occupy a small and constrained region of the broader fluorinated chemical space, providing the chemical basis for the empirical stability of the Effective Fluorine Conversion Factor (EFCF).